SMART CITY TRAFFIC FLOW ANALYSIS REPORT

1. ABSTRACT

This project focuses on analyzing urban traffic patterns using Python and Power BI. A synthetic dataset of 100,000 records was generated using Faker to simulate real-world traffic scenarios. The system identifies congestion trends, peak hours, and external factors like weather and incidents, enabling data-driven smart city planning.

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2. INTRODUCTION

With increasing urbanization, managing traffic efficiently is a major challenge. Smart city solutions rely on data analytics to monitor and optimize traffic flow.

This project implements a traffic analysis system using Python for data generation and analysis, and Power BI Desktop for visualization. It simulates real-world traffic behavior and provides insights into congestion and mobility patterns.

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3. OBJECTIVES

- To generate realistic traffic data using Python
- To analyze traffic trends and congestion levels
- To identify peak traffic hours
- To study the impact of weather and incidents
- To build an interactive dashboard using Power BI

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4. SYSTEM MODULES

4.1 Data Generation Module

- Generates 100K records using Faker
- Simulates uneven traffic patterns

4.2 Data Analysis Module

- Performs trend and correlation analysis
- Identifies peak hours and congestion

4.3 Visualization Module

- Interactive dashboards in Power BI
- KPI tracking and filtering

4.4 Feature Engineering Module

- Peak hour detection
- Traffic categorization

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5. FEATURES OF THE SYSTEM

- Realistic synthetic dataset
- Uneven traffic patterns (peaks & dips)
- Traffic trend analysis
- Weather & incident impact analysis
- Interactive Power BI dashboard
- Feature engineering for insights

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6. USAGE OF THE SYSTEM

The system starts by generating a dataset using Python. The data is then analyzed to extract meaningful insights such as peak hours and congestion trends.

The processed dataset is imported into Power BI, where interactive dashboards are created. Users can filter by city, weather, and road type to explore traffic behavior dynamically.

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7. DATASET DETAILS

The dataset includes:

- Datetime, Hour, Day, Month
- City, Road Name
- Vehicle Count, Avg Speed
- Congestion Level
- Weather, Road Type
- Incident

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8. ADVANTAGES

- Realistic traffic simulation
- Helps in smart city planning

- Interactive and user-friendly dashboard
- Scalable for larger datasets
- Demonstrates real-world analytics workflow

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9. LIMITATIONS

- Synthetic data (not real-time)
- No live traffic integration
- Basic analysis (no deep AI model)

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10. FUTURE ENHANCEMENTS

- Add real-time traffic APIs
- Implement machine learning prediction
- Build web dashboard using Flask
- Add geospatial mapping
- Integrate IoT traffic sensors

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11. CONCLUSION

This project demonstrates how Python and Power BI can be used to analyze and visualize traffic data for smart city applications. It provides valuable insights into traffic patterns and lays the foundation for advanced analytics and predictive systems.